

Lab 3: Demonstrate Tools and an MCP Design Boundary

Business Innovation with AI Agents · TGS-2020503395 · 40 minutes

Objective

Use file and terminal tools for a grounded calculation, then design a least-privilege MCP connection without live credentials.

Deliverable

A calculated service analysis, tool trace, MCP trust register and failure record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 2 task-contract pattern.

Procedure

Step 1. Inspect mock-tickets.csv and data-dictionary.md. Confirm the data is fictional.

Step 2. Ask Hermes to calculate ticket count, median resolution time and breach rate using a terminal or code-execution tool.

```
hermes chat --toolsets file,terminal
```

Step 3. Require the agent to save analysis.md and include the calculation method.

Step 4. Run independent-check.py and compare its result with the agent output.

```
python3 independent-check.py
```

Step 5. Complete mcp-design-template.md for a hypothetical CRM server: transport, scopes, read/write tools, trust tier and approval rules.

Step 6. Mark every write-capable MCP tool as approval-required and exclude customer secrets from prompts and logs.

Step 7. Record one failed or blocked tool call and identify whether the owning boundary was schema, permission, execution or evidence.

Step 8. Run the bundle verifier.

```
python3 verify.py
```

Acceptance test

Independent calculations match the reported metrics, every claim points to the mock file, and the MCP design exposes only necessary capabilities.

Evidence checklist

- ☐ Commands/configuration with secrets redacted
- ☐ Screenshot or file evidence
- ☐ Acceptance test result
- ☐ Failure and resolution
- ☐ Reflection and improvement

Safety and cleanup

- Use fictional data only.
- Never submit .env, tokens, keys or personal information.
- Stop classroom gateways and remove recurring jobs after evidence capture.